Received: 25 April 2018 | Revised: 22 March 2019 | Accepted: 8 April 2019

DOI: 10.1002/isd2.12098

RESEARCH ARTICLE

WILEY

Applied business process management: An information systems approach to improve service delivery in public hospitals of low- and middle-income countries

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Abstract

South African public hospitals are faced with many challenges comprising inefficient processes, lack of digitalisation, severe austerity measures, and the rising burden of disease. Irrespective of the many existing initiatives in place, a critical challenge is the shortage of necessary resources required to fund, implement, and sustain new interventions. This deprives public hospitals from the efficiencies and quality of service experienced in private hospitals or other business entities. Through a case study strategy, this paper has abstracted two practice contributions. First, it describes and prescribes a low-cost approach in which business process redesign was applied in a public hospital through a partnership between the hospital and a higher learning institution. The initiative involved third year students majoring in Information Systems performing, as part of their class project, process analysis at the hospital at no cost to the hospital other than their employees' time. The second contribution is a priority list of redesign and lean management principles that were abstracted from these initiatives. These include improving staff training, distributing and adhering to up-to-date standard operating procedures, performing tasks in parallel rather than in series, reducing handovers by avoiding certain tasks, incentivizing use of existing mobile software by doctors, developing low-cost task automation solutions such as Microsoft Office applications, redesigning and standardizing forms, capturing data at source electronically rather than later, redesigning physical spaces, and improved key performance indicators and reporting. The study confirms the usefulness of this collaborative business process redesign approach for public hospitals as stakeholders got relevant information and practical, low-cost, creative solutions with the potential to improve efficiencies and subsequently service delivery.

KEYWORDS

business process management, health systems, service delivery, business process re-engineering, business process redesign, lean management

1 | INTRODUCTION

South Africa, a developing country, is more specifically classified as an upper middle-income economy (Stevens, Mascarenhas, & Mathers, 2009). In 2013, it spent 8.9% of its GDP on healthcare, like the Organization for Economic Co-operation and Development (OECD) global average of 8.9% (OECD, 2015). Yet despite that spend, life expectancy in South Africa in 2013 was calculated as 57 years, much lower than the OECD average of 81 years (OECD, 2015). South Africa's health system is distinctly divided into private and public, on the basis of the financial capacity of the healthcare consumer. Private healthcare is mainly funded by insurance schemes and patients who can afford to buy health services. It serves only 20% of the population. The overburdened public healthcare relies on government, competing for a portion of limited funds, and assists 80% of the population with 20% of the resources of the private system (Wallis, Garach, & Kropman, 2008; Weimann & Stuttaford, 2014), subsequently fuelling extreme disparity. Other well-documented challenges are inefficiencies, a lack of information, corruption, severe austerity measures, and rising costs of care (Chopra, Daviaud, Pattinson, Fonn, & Lawn, 2009; Weimann & Stuttaford, 2014). Therefore, the South African healthcare system urgently needs reform as proposed by the National Health Insurance (NHI). Health service quality standards and norms have been developed in the form of the National Score Standards (NCS), but few public healthcare facilities audited have been found to comply with them (Mayosi et al., 2012).

There are obstacles and inefficiencies in the process flows of South African public hospitals (Stuart-Clark et al., 2012; Weimann & Stuttaford, 2014). Studies have reported frustration expressed by patients at the long waiting time in queues at emergency departments before being treated (Armony et al., 2011; Weimann & Stuttaford, 2014). This shortage of inpatient beds ultimately affects the flow of admitted patients: patients who are forced to wait until beds become available (Eitel, Rudkin, Malvehy, Killeen, & Pines, 2010). Another inefficiency stems from the lack of a proper record-keeping system and insufficient electronic systems, which needs strengthening to improve service delivery (Stuart-Clark et al., 2012). Other inefficiencies are imbedded in the high amount of bureaucracy sketched in the procurement process of South African public hospitals. Information systems are not designed to consider the conditions and business processes of the users of these systems. This results in most health information system implementations in developing countries being partial or complete failures (Johnston, Jali, Kundaeli, & Adeniran, 2015). Process innovation, such as the use of mobile devices within health care has been mentioned as one of the essential components of digital health, innovation, and digital ecosystems (Iyawa, Herselman, & Botha, 2017).

One approach to improve hospitals is applying management tools and techniques that are well established in the industry and have a track record of enabling business efficiency. Successful examples can be seen in lean management and business process management (BPM; Clancy, 2009; Kim, Spahlinger, Kin, & Billi, 2006). Yet BPM adoption has been described as an uphill battle because most businesses are organised by function (Tucci, 2007). This is even more eminent in public sector organisations that tend to have a hierarchical culture (Cameron & Quinn, 2006). The process perspective requires collaboration, hand-holding, sharing, and negotiation across hierarchies that are often not incentivised; hence, external support is often needed (Tucci, 2007). Therefore, although public hospitals in middle-income countries would benefit from BPM (Weimann & Weimann, 2017), they are also unable to afford the initial investment needed for more conventional BPM project approaches.

The literature does not show a rigorous comprehensive framework to address complex and vast healthcare environments of developing countries (Künzle & Reichert, 2011), although there are attempts to advance models in developed countries (Cleven, Winter, Wortmann, & Mettler, 2014). Therefore, this research takes a pragmatic stance and describes and prescribes a low-cost redesign approach through abstracting redesign principles successfully applied in a public hospital of a middle-income country.

2 | LITERATURE REVIEW

A business process is a set of activities or logically related tasks performed to deliver value to customers or to fulfil other strategic goals (Trkman, 2010). A further important characteristic of a business process is that it crosses organisational boundaries (Davenport & Short, 1990). Hence, often employees lack the power to change business processes without external support (Malhotra, 1998).

BPM is a management practice that integrates business process knowledge and information technology (IT). It is a recent advance in harnessing business processes as exploitable assets (Niehaves, Poeppelbuss, Plattfaut, & Becker, 2014; Thompson, Seymour, & O'Donovan, 2009). BPM maturity models can be used to characterise how organisations are currently managing their processes (their level of BPM maturity) and can assist organisations in developing BPM strategies and guiding their process efforts (Harmon, 2015).

BPM incorporates many prior approaches to business process change. These prior approaches come from three traditions (Harmon, 2015): first, the quality control tradition that includes total quality management, Lean and Six Sigma, which focus on quality and continuous process improvement; second, the management tradition, which stresses innovation and changing business (mostly through process change) to achieve competitive advantage and includes business process reengineering and business process redesign. The third tradition, IT, addresses software automation of processes. We now briefly describe some of these approaches including their relevance in public hospitals.

Lean focusses on a holistic approach involving all participants in the organisational hierarchy to continually improve all processes and functions and reduce waste (Weimann & Weimann, 2017, p. 55). The eight forms of waste in hospitals are as follows: (a) over-processing, (b) unnecessary movements, (c) corrections, (d) transport, (e) waiting, (f) stock, (g) overproduction, and (h) inadequate use of staff. Lean operates through the improvement of standardised processes and activities, aiming to reduce nonvalue-adding activities and eliminate waste (Harmon, 2014; Womack, Byrne, Fiume, Kaplan, & Toussaint, 2005). Lean does not require an external project team and is increasingly being applied within the healthcare sector and, for example, has been used to improve theatre productivity (Meredith, Grove, Walley, Young, & Macintyre, 2011). Kanban is a flow-based Lean method that originated in manufacturing (Lei, Ganjeizadeh, Jayachandran, & Ozcan, 2017), is used in managing software development (Ahmad, Kuvaja, Oivo, & Markkula, 2016), and has more recently been applied in hospitals to minimise personnel errors and in managing medical supplies in wards (Regattieri, Bartolini, Cima, Fanti, & Lauritano, 2018).

Outside-In (Gulati & Gilbert, 2010; Towers, 2010) is one of the newest approaches to business process change. It operates with the understanding that the end users experience is the process and has a critical or almost obsessive focus on the customer. It aims to remove or improve every end user touch point, mapping moments of truth, breakpoints, and business rules and striving to achieve success through consistent use of process to effectively remove processes. There are limited cases of this approach in the research literature.

Business process reengineering, which was trendy in the 1990s, is associated with drastic process change projects. In contrast, business process redesign, which has succeeded it and which we will refer to as redesign in this paper, is less radical (Harmon, 2014; Limam Mansar & Reijers, 2007).

2.1 | BPM challenges

BPM challenges are numerous, and in hospitals in developing countries, the challenges appear greater. The South African public healthcare system suffers from inefficient health service delivery (Benatar, 2004). However, the system also experiences a lack of resources, including a lack of skilled human resources, poor management, overcentralised decision-making, and a lack of transparency and accountability (Benatar, 2004), which all present barriers for BPM. For example, public sector organisations in Brazil, a middle-income country, experience low BPM maturity (Valenca, Alves, Santana, de Oliveira, & Santos, 2013). Government organisations have characteristics that hamper BPM adoption, such as periodic changes in government leaders, which results in discontinued BPM initiatives, low flexibility, a lack of innovation, and rigid hierarchical structures (Valenca et al., 2013). The BPM core element framework (Rosemann & vom Brocke, 2015) identifies the six essential elements of BPM as strategic alignment, governance, methods, information systems, people, and culture. Organisations grapple with all six.

In terms of strategic alignment, organisations have struggled with insufficient alignment of BPM efforts and organisational strategy (Malinova, Hribar, & Mendling, 2014). BPM governance defines rules that address control, process ownership, and BPM strategic alignment (Santana & Alves, 2016). Generally, governance of BPM is lacking (Bandara, Indulska, Chong, & Shazia, 2007; Buh, Kovačič, & Indihar Štemberger, 2015). Appropriate governance can reduce corruption: In South Africa, the siphoning of scarce resources due to corruption has gravely affected health (Agyepong et al., 2017).

In terms of culture, a focus on team domain competences poses a challenge (Jurczuk, 2016), which pervades public hospitals. For example, cultural changes within public sector organisations in Brazil were seen to be critical due to their bureaucratic structures and departmentalised culture, which negatively impacts BPM (Valenca et al., 2013). Also, in India, a middle-income country, cultural characteristics of high-power distance, collectivism, and low uncertainty avoidance have obstructed BPM and resulted in highly informal BPM governance (Jayaganesh & Shanks, 2009).

People can be a barrier, for example, when there is insufficient understanding of organisational processes (Sadiq, Indulska, Bandara, & Chong, 2007) and a lack of BPM expertise, which poses a major threat for public sector BPM initiatives (Valenca et al., 2013). Most African countries also suffer from a major shortage of healthcare workers mostly attributed to weak educational systems and inadequate investments in training (Agyepong et al., 2017) and investment in BPM.

IT can be a challenge especially when the organisation, as in the case of public hospitals, has insufficient IT investment and support (Buh et al., 2015). There has been a call to improve health information systems in South Africa (Mayosi et al., 2012). Low- and middle-income countries face multiple challenges when introducing much needed IT systems. IT literacy and financial resource constraints are some of the major contributors (Avgerou & Walsham, 2018; von Holdt & Murphy, 2007).

Finally, methods can prove a challenge, due to a general lack of BPM methodology and standards (Bandara et al., 2007). It is the latter challenge that we address in this paper.

2.2 | Business process redesign

Ideally, redesign of a business process, either decreases the time to perform the business process, decreases the required cost of executing the business process, improves the quality of the service delivered or enhances the ability of the business process to react to variation. Redesign is still considered more art than science, researchers have published redesign best practices with the most popular best practice performed by practitioners being "task elimination" (Limam Mansar & Reijers, 2007). Redesign methodologies such as BPTrends are developed for redesigning relatively large processes by a multidisciplinary project team (Harmon, 2014) and typically involve high levels of technology, such as computer simulations, in the redesign (Palma-Mendoza, Neailey, & Roy, 2014). The six stages in most redesign methodologies include strategic vision and business understanding, selection of target for redesign, understanding the as-is business process, designing the to-be process, implementation, and evaluation (Palma-Mendoza et al., 2014). Our focus in this paper is on the two stages of understanding the as-is and designing the to-be business process. Researchers have acknowledged that getting from the “as-is to the to-be” has the least clarity and has been referred to as invoking the ATAMO (And Then A Miracle Occurs) procedure (Limam Mansar & Reijers, 2007). We will refer to this loosely as redesign.

3 | METHOD

The philosophical stance in this research adheres to Pragmatism. Pragmatism provides a mechanism to help explain why things either do or do not work by asking the correct questions and getting empirical answers to the questions (Baskerville & Myers, 2004). Pragmatism is concerned with action and change and the interplay between knowledge and action with the intended goal of intervening in the world and not merely observing (Goldkuhl, 2012).

The organisational settings or context of ICT use within sectors, countries, or regions differ. For example, countries with different public administration traditions practice e-government differently. Hence, there is a need for studying more socially embedded processes particularly within developing countries (Avgerou, 2010) such as in this paper, which follows an embedded single case study strategy (Yin, 2011). Four embedded cases within one public sector hospital are described, and from them, we distilled prescriptions.

There is also a proposed disconnection between practitioners and academic scholars. A solution suggested to resolve this disconnect is to extract or uncover rules used in practice. This type of research is referred to as the “extracting case study” approach and has produced valuable contributions such as the Kanban-system applied in continuous process improvement (Gregor, 2009; van Aken, 2004). Mueller and Urbach (2017) refer to five types of reasoning: inductive, deductive, abductive, retroductive, and theorizing through design and action. This research uses the latter reasoning, which involves reflecting on what has been done and the abstracting of design principles, which produces theory for design and action (Gregor, 2009). The subsequent technological rules are defined as midrange theories of practice, only valid for a certain domain (Gregor, 2009). Gregor (2009) references Davenport and Short's (2009) abstraction of the general method of business process redesign from 19 case studies as an exemplar of this type of research. As our contribution related to business process redesign, it therefore seemed fitting that we follow this approach.

3.1 | The course and analysis of redesign steps

In the redesign intervention approach, student teams perform the redesign as a class project in a BPM course for third-year students in the Information Systems (IS) major. The BPM course runs over 12 weeks, and the project is done over 10 weeks. Each team has four to five students in it. Each team is assigned a postgraduate student as a mentor, and each team chooses a business process to redesign as part of the in-course project. Two authors are teachers that are responsible for introducing students to the necessary skills, concepts, and tools required for the project. We are also responsible for the overall course/project administration including assessing and providing feedback to students on the various project deliverables. The assignments of the IS students are evaluated at two stages using a standardised template by a panel of academic staff (UCT) and by the process owners.

The redesign steps and procedures were developed over the years and are refined by the course lecturers annually since 2009. The steps are a subset of steps mostly based on the BPTrends methodology (Harmon, 2014). We experienced these steps to be easily performed by student teams; process owners have found them to add the most value to the redesign effort. The focus was on a short time frame (the student project) and using technology that could be easily learnt by students.

3.2 | The hospital intervention

From 2015 to 2017, eleven processes were redesigned at Groote Schuur hospital. For this paper, we analysed and evaluated four of these projects as our four cases. The three authors of this paper participate annually in the hospital intervention as lecturers on the BPM course and coordinators of the student projects. One of the authors, based at the hospital, was responsible for managing the interaction between students and various stakeholders at the hospital. The researchers annually obtained ethics approval from the university and the hospital. The processes chosen focused on various daily operations that were known to be problematic. Prior to students entering a hospital situation, students agree to nondisclosure of any confidential information obtained as part of the project and that the intellectual property for the redesign would reside with the university.

Groote Schuur Hospital (GSH) is an academic tertiary hospital in the Western Cape province of South Africa that provides 975 inpatient beds and state-of-the-art clinical treatment. It is internationally renowned for the first human-to-human heart transplant that took place in 1967 (Mars & Seebregts, 2008; Patel, 2014; Western Cape Government, 2014a). The hospital is committed to providing “access to patient-centred, quality care, by adopting a theme of leadership, innovation, and change” (Patel, 2014).

GSH was chosen for this study because of its innovation program that seeks to inventively improve hospital processes. The initiative aims to tackle eight challenges through the creation of a hospital innovation hub. They are as follows: tracking and communicating, patient records and notes, more efficient entry and exit, reducing waiting time, sustaining a culture of care and dignity, improving care for specific patient groups, working better with district health services, and boosting volunteer resources (Patel, 2014).

Public tertiary hospitals are typically very busy. It is therefore vital that the project does not interfere with daily operations and responsibilities of the hospital employees. It is also imperative to adhere to the various ethical and professional regulations.

GSH falls under the Western Cape Government Department of Health, which has released its Healthcare 2030 vision stated as "Access to person-centred, quality care" (Western Cape Government, 2014b, p. 60). The vision includes a commitment to have all health facilities provided with basic information systems and to have auditable and effective monitoring, reporting, and evaluation of performance. The department also commits to develop internal expertise in "BPR strategies such as Lean" (p. 82).

3.3 | Data collection and analysis

The qualitative documents analysed for this study include evaluations completed by staff in the hospital (process owners who would apply the methods) and the actual assignment reports completed by student groups. The four cases with the eight documents analysed are listed in Table 1. We analysed the student reports to generalise and extract best practice redesign principles (the second contribution). These best practices were distilled to reduce waste in the hospital according to the lean management principles, improving outcomes and efficiencies. We then analysed the evaluations to assess the redesign principles and the effectiveness of the actual redesign described.

4 | RESULTS

In this section, we describe and prescribe a low-cost redesign approach that we applied. The value of the contribution is describing how these redesign tasks can be used within a public hospital in a developing country. Figure 1 shows the steps of the redesign. The steps followed are based on the BPTrends redesign method (Harmon, 2014). We followed most of the recommended steps; however, two changes were made.

- First, we omitted performing costings for all activities. This was seen to be less relevant as public hospitals are not focused on profit but rather on patient care. Over the years, most organisations did not appear to value detailed costings but preferred that students focus on innovative solutions.
- Second, we included an extra step where reports were specified. This step was very valuable as it allowed the organisation to visibly see how their processes could be managed in the future. Organisations found this step valuable and enjoyed a tangible outcome to the project in addition to the report.

To reduce the length of this paper, we omit examples of some steps such as the business case, the project plan, and the organisational diagram but refer readers to the BPTrends templates (Harmon, 2014) for these. The different elements of the assignment are now described with examples for the different processes.

TABLE 1 Data sources

Case	Year	Process description	Evaluator(s)	Report	Evaluation
B	2016	Medical ward discharge process	Bed Flow Manager	B1	B2
E	2016	Outpatient department medical coding	Outpatient Medical Manager	E1	E2
D	2015	Bulk procurement process	Head of Pharmacy	D1	D2
F	2015	Internal procurement process	Director Finance and Information Management	F1	F2

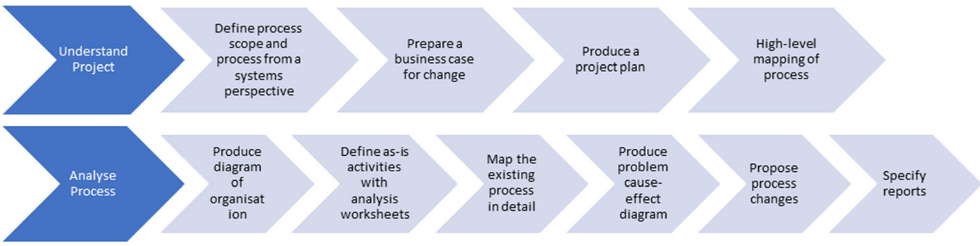

FIGURE 1 Low-cost redesign steps followed

The first phase of the process analysis is conducted through onsite meetings with three different process stakeholders. This allows the teams to gather a preliminary understanding of the process. As an outcome of these meetings, students draw and discuss a scoping diagram that assist in understanding the process from a systems perspective. They also produce a business case for changing the process and a high-level mapping of the business process. The outcome of the first phase is then presented to relevant stakeholders and course instructors for input and guidance.

The second phase requires student teams to perform another round of site visits, this time with the aim of understanding on a deeper level the problems and risks with the existing process as well as identifying the causes of process problems. Suggestions on process redesign and specified operational and management reports/dashboards for process management are further developed. Student findings from both phases are compiled into a report that is submitted for marking and handed to process owners with the expectation that recommendations will be given serious implementation consideration. The relevant activities and report elements are now described.

4.1 | Define process scope and the process from a systems perspective

To initially understand the process and its problems, the students sketch a scoping diagram. In Figure 2, we display a scoping diagram for the *bulk procurement process*. This diagram represents high-level elements of the process focusing attention on six generic process problems, namely, input problems, output problems, problems with controls, problems with enablers, process flow problems, and day-to-day management problems (Harmon, 2014). The solid line in the diagram also depicts the process boundary and what is in or out of scope for the redesign.

At a high level, this process was described by four stages: receiving stock levels on JAC (the internal pharmacy stock management system), placing orders with suppliers, receiving orders, and generating invoices. Problems identified are now described. Input problems included quoting incorrect stock to suppliers and output problems included delayed payments by the finance department. Control problems related to standard operating procedures (SOPs) not being distributed and enabler problems related to insufficient training of assistants, wastage due to expired stock, limited storage, and the slow process for ordering using the existing centralised ordering application (IPS). The use of IPS for sourcing buyout quotes was a new requirement request by treasury and enforced in September 2015 when the process was analysed.

4.2 | Produce problem cause–effect diagram

A problem cause–effect analysis was then done to understand causes of the current process problems. Figure 3 shows a problem cause–effect diagram of the *medical ward discharge process*.

Five causes were identified of the slow ward discharge process: discharge plans not being followed, solely paper-based systems, no process manager or supervisor, minimal use of appropriate IT, and missing standardised procedures (SOPs) to fill empty hospital beds. The paper-based

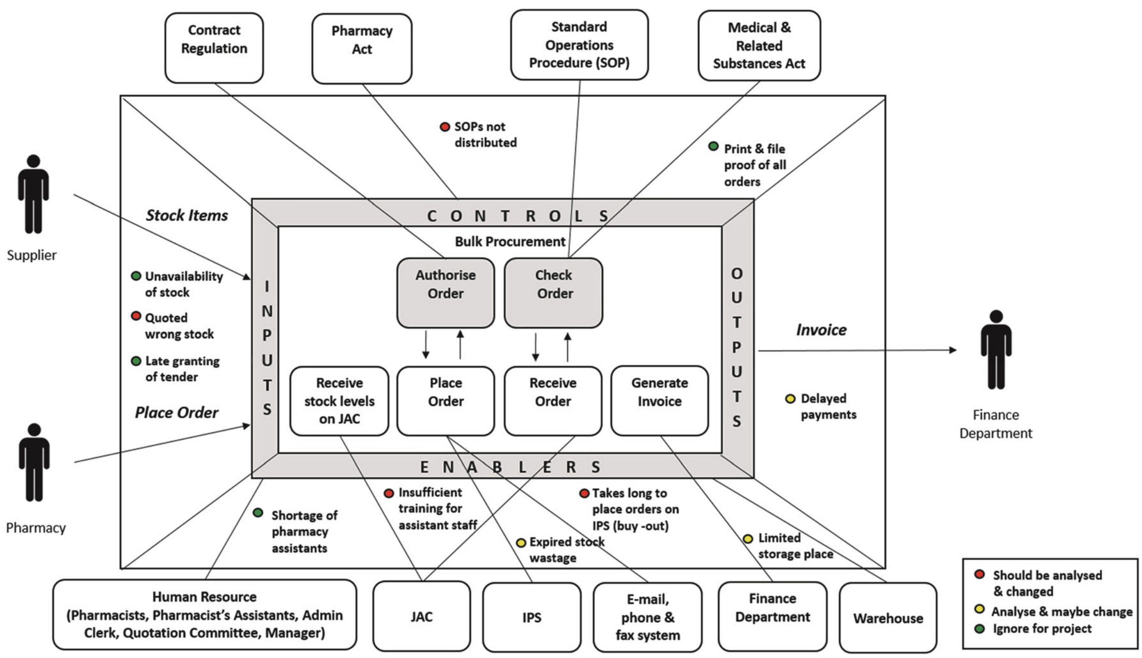

FIGURE 2 Scoping diagram for the bulk procurement process (D1)

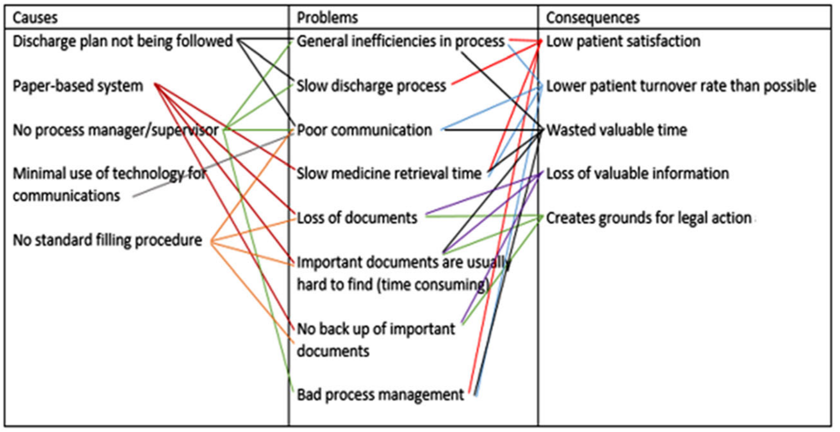

FIGURE 3 Problem cause–effect diagram of the medical ward discharge process (B1)

system resulted in slow medicine retrieval times and loss of documents, it was also time consuming to find important documents and no back-up system of important documents was in place. The visible consequences in the ward included the following: low patient satisfaction with a potential risk of official complaints and legal actions against the hospital, low patient turnover during the day with resultant overcrowded intake wards and emergency departments, patients are often discharged late in the evening, and time wastage for staff, patients and family members as information is lost with resultant risks to patient treatment.

4.3 | Map the existing process

The teams needed to map or model the process twice: first, at a high level and after feedback at a more detailed level. Figure A1 shows the model of the prior bulk procurement process modelled using Business Process Model and Notation (BPMN). BPMN is a standard modelling language that is taught in the BPM course. Each task in the process is indicated using a smoothed rectangle, and decisions are indicated with diamonds. Each column represents a role, and all tasks performed by that role are therefore placed in that column. The diagram shows where tasks are handed over between roles as well as communicated to outside roles such as the contract supplier and the buyout supplier. The analysis showed that most of the work was done by the pharmacist, followed by the pharmacist's assistant. The diagram allowed students to identify substantial work in following up on suppliers being late with deliveries or quotes. Conflicts of interest were additionally identified: In this case, the same person who unpacked stock was also checking on the unpacking of stock.

4.4 | Propose process changes and 10 best practices

Students then propose solutions that are now described for all cases, and the recommended changes are abstracted and summarised as 10 best practices, presented in Table 2.

For the medical ward discharge process, the main cause of inefficiency was identified as the sequential way activities were performed. Hence, it was proposed that many activities could happen in parallel and the sequence of some tasks could change with resultant reduction in delays and waiting. This change is abstracted as *performing tasks in parallel rather than in series*. It was suggested that the following three tasks happen in parallel: preparing the patient for discharge, writing up the discharge letter, and obtaining take-out medicine from the pharmacy.

Labour intensive manual tasks were highlighted such as runners transferring inputs and outputs between wards and pharmacies. These tasks increase human error (corrections) and delays (waiting) and unnecessary movement. A recommendation was to use the shoot system[1] whenever possible. This is abstracted as *a low-cost task automation solution*.

For the *internal procurement process* a Lean and Outside-In approach was proposed. A major problem identified was long wait times between the initial requisition and final procurement. The suggested Lean approach included the use of a standardised Excel form to reduce the amount of waiting and unnecessary movement that form part of physically moving a paper requisition. This was conceptualised as developing low-cost automation solutions and redesigning and standardizing forms. An advantage of the form is input and data integrity controls that can prevent errors before they occur and remove that responsibility from the administrative clerks. The second solution proposed focused on delays due to the requisitioner needing to choose between different quotes. The form was amended with an opt-out of having to choose quotes, by preselecting the lowest cost quote or a contract supplier. This approach was conceived as "reducing hand-overs by avoiding certain tasks" so that the final step of confirming the supplier could be avoided.

[1]Shoot system is a pneumatic tube system within a hospital to transport lab samples and files.

TABLE 2 Redesign best practices per case

10 best practices/case	Medical ward discharge	Internal procurement	Bulk procurement	Outpatient department medical coding
Improving staff training			X	X
Distributing and adhering to up-to-date SOPs			X	
Performing tasks in parallel rather than in series	X			
Reducing hand-overs by avoiding certain tasks		X		
Incentivizing use of existing mobile software by doctors				X
Developing low-cost task automation solutions such as Microsoft Office applications	X	X	X	
Redesigning and standardizing forms		X		X
Capturing data at source electronically rather than later				X
Redesigning physical spaces			X	
Improved key performance indicators and reporting			X	

Whereas IS students might be criticised for focusing on the latest expensive IT solutions, the teams were asked to recommend low-cost solutions. For the bulk procurement process, a multipronged approach to process change with minimum IT was suggested. First, the pharmacist's assistant rotational programme was identified as not being followed and being exacerbated by the lack of distributed SOPs. Reinstating the rotational programme (*improving staff training*) and *distributing up-to-date SOPs* would improve the efficiency of the entire process, reduce the amount of expired stock and wasted time. It was identified that stock was not stored in the correct places due to space constraints leading to confusion when picking stock and hence wasting time. The gate to the quarantine room where stock is delivered was also identified as not being wide enough to allow entry for some larger deliveries, resulting in stock being placed in the entrance area of the main bulk store. This makes it hard to move and wastes time. A proposal was to widen the shelf depth and a resolution of the gate problem was recommended (redesigning physical spaces). Also, better key performance indicators (KPIs) were recommended to monitor current stock enabling the timely ordering of new stock.

The prior bulk procurement process had three measures in place (number of out of stock items, number of payments made outside of 30 days and value of expired stock). These were not checked very closely and were insufficient to evaluate the process. It was also noted that supplier orders were not recorded. Hence, late deliveries could not be pre-empted leading to stock-outs. The team recommended a low-cost Excel workaround with embedded macros to record orders, monitor deliveries, enable automatic supplier reminder emails and report on their KPIs (improved KPIs and reporting, developing low-cost task automation solutions).

For the capturing and reporting of *Medical codes in the Outpatient Department*, students identified that recording of ICD10 codes[2] on paper forms and physically moving forms between the departments and reception leads to lack of information, incorrect capturing, incorrect and delayed reporting, and severe reporting lags that introduced further errors. The information unit was performing most of the capturing as delayed batch processing. There were many missing and incorrect codes partly due to poor doctor handwriting. The students recommended ensuring sufficient and better trained clerks for on-site capturing (improving staff training), incentivizing doctors to use the Clinicom[3] system, using available ICD 10 coding applications on their smartphones (incentivizing use of existing mobile software by doctors, capturing data at source electronically rather than later) and redesign of the patient diagnosis form (redesigning and standardizing forms).

In summary, the following best practices distilled from these four cases are good examples of low-cost process changes that would result in more efficient and effective services, reduce waste, and improve outcomes.

- Improve staff training (reduce inadequate use of staff)
- Distribute and adhere to up-to-date SOPs (reduce waiting)
- Perform tasks in parallel rather than in series (reduce waiting)
- Reduce and standardise handovers (reduce overprocessing and waiting)
- Incentivise use of mobile software by doctors (reduce waiting and correction)
- Develop low-cost task automation solutions such as Microsoft Office applications (reduce unnecessary movement, waiting, and correction)
- Redesign and standardise forms (reduce corrections)

[2]ICD 10 is an international classification of diseases.

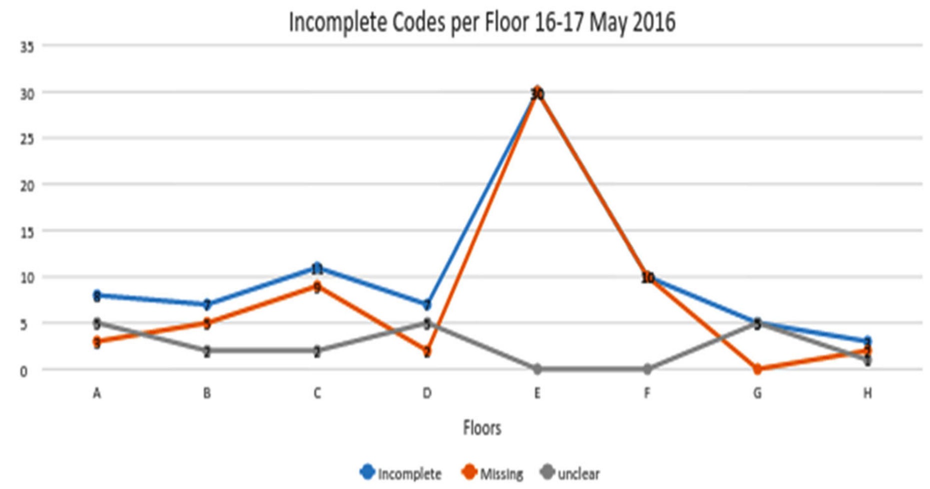

FIGURE 4 Operational report for the medical coding case (E1)

- Capture data at source electronically rather than later (reduce overproduction)
- Redesign physical spaces (reduce stock and waiting)
- Improve key performance indicators and reporting (avoid over-processing and waiting)

4.5 | Specify reports

The teams were asked to propose and specify appropriate operational and management reports to monitor the efficiency and effectiveness of the business processes and track the impact of the recommended process changes. Figure 4 displays a report proposed for *the medical coding case*. The report indicates incomplete and missing codes on specified dates for each floor (A–H) of the hospital. The team recommended that the report should be compiled more frequently. In cases of noticeable anomalies, as is the case for Floor E in Figure 4, the information desk can then resolve problems timeously.

Figure 5 presents dashboard report visualisation techniques. The dashboard is divided into four subscreens relevant to the bulk procurement process. This report displays metric fluctuations across different months in a year. Students also described the data sources for the reports. Ideally, the baselines or thresholds beyond which anomalies can be flagged for follow-up should also have been shown. This would have made the graphs more informative.

4.6 | Evaluation of process redesign

In this section, comments by stakeholders on the value they got from the process redesign were analysed. Stakeholders noted the quality of the analysis done in terms of relevance, insightfulness, and recommendations. The stakeholders found their own investment of time worthwhile as they got relevant information and output:

> *The project was very informative and lots of details went in. (E2)*

> *Highly valuable information, especially the relatively short time frame the team was given to obtain insight of a new field. (B2)*

Even though the students were not from a medical discipline and were young, their fresh perspective was appreciated, and they were found to have high problem-solving ability and insight:

> *Well done! The team obtained excellent insight into a very complex and demanding hospital management issue … Excellent problem-solving approach. (B2)*

> *The students had a very limited knowledge of the process prior to the project, however they were thorough in their research. (D2)*

The recommendations were found to be relevant, related to existing challenges, practical, low-cost, creative, and with potential to improve efficiency.

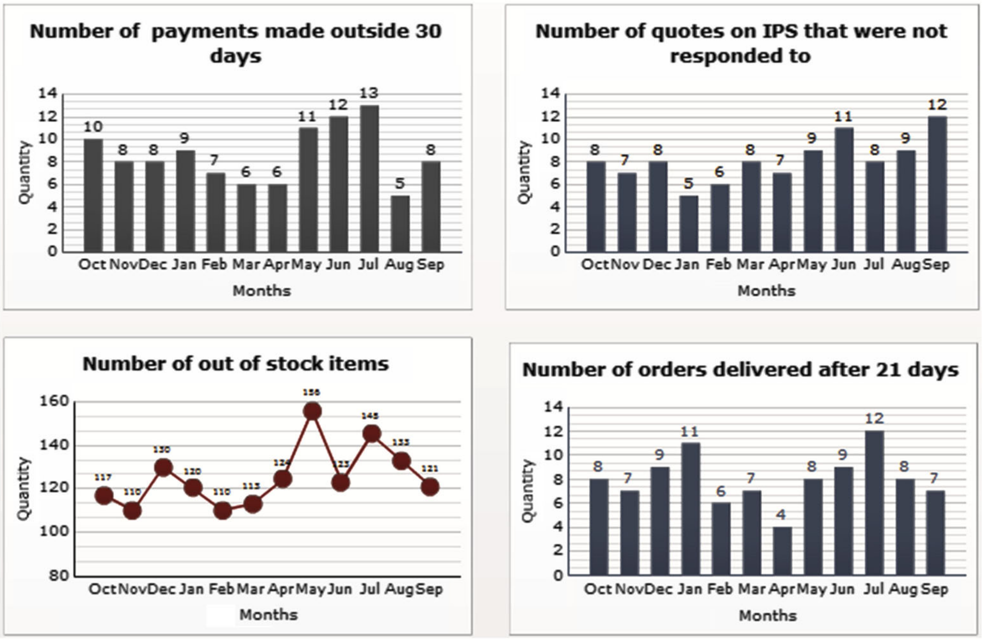

FIGURE 5 Managerial report for the bulk procurement process (D1)

> *They listened to what our challenges were and strove to provide solutions to those challenges. (D2)*

> *The proposed recommendations could allow the activities in the discharge process to happen in parallel, which will reduce the total process time and improve communication between employees, patients, and transport. (B2)*

Yet there were challenges. In some cases, staff in the hospital were reluctant to share process problems with the students.

> *As a team we kind of felt as if some people we interviewed were holding back on issues like the problems they faced with the current system, making it hard for us to really put in place ways to improve a seemingly working system. (D1)*

In addition, some recommendations could not be taken up after discussing the feasibility with the involved staff:

> *The students recommended that staff be rotated for the random daily stock checks. This was discussed with the supervisors and not implemented as they felt it would be difficult to manage the daily movement of staff and it may lead to more discrepancies if staff were not familiar with the items being counted. (F2)*

In some cases, stakeholders have implement the suggested changes:

> *The students recommended using Excel to mail suppliers to remind them that an order is due. We have started a spreadsheet to record dates when orders are sent to the supplier. SharePoint can be used to create automatic e-mail to suppliers from this spreadsheet. (D2)*

> *We will review our SOPs. (E2)*

Yet, there were concerns regarding available resources:

> *My concerns really are that to implement the recommendations we would need extra resources. (E2)*

> *The most common challenges are the buy-in of the staff and the capability of the current infrastructure. (B2)*

However, even where recommendations were not considered feasible, the experience was a unique opportunity to reflect on operations.

> *An external view on systems used in daily practice usually makes one take a closer look at practices to determine if there are any efficiency gains to be made. The report created an awareness of potential risks in the current system e.g. stock counts and way in which training is conducted. Although all recommendations were not implemented, it was a useful exercise as the report has stimulated the thinking of using technology to improve management of stock outs. (D2)*

In summary, the analysis of business processes in the hospital was generally seen by the involved staff as a useful exercise and recommendations were implementable. We therefore see value in following this approach in other public hospitals in developing countries.

5 | DISCUSSION

The literature describes the South African public healthcare system as suffering from inefficient health service delivery (Benatar, 2004; Weimann & Stuttaford, 2014) and described existing challenges. In this study, we found evidence of challenges in terms of all the six core elements of BPM (Rosemann & vom Brocke, 2015). We also note that many of our 10 best practices address these challenges.

In terms of strategy, organisations are known to struggle with aligning BPM efforts and organisational strategy (Malinova et al., 2014). Although the healthcare 2030 strategy committed to effective monitoring, reporting and evaluation of performance, one of the greatest challenges the student teams had was collecting baseline data of performance. There was also no internal process expertise. This speaks to clear misalignment with strategy.

Public South African healthcare has been attributed with poor management, corruption, and a lack of transparency and accountability (Benatar, 2004; Weimann & Stuttaford, 2014), a hierarchical culture with overcentralised decision-making and bureaucracy (Benatar, 2004; Cameron & Quinn, 2006; Johnston et al., 2015). In terms of management and culture, in this study, we noted a lack of transparency, overcentralised decision-making and bureaucratic processes. Students felt that staff were keeping information confidential and there were many approval delays in processes that were deemed unnecessary. Practice 2, “distributing and adhering to up-to-date SOPs,” and practice 10, “Improved key performance indicators and reporting,” improve transparency and accountability. Practice 4, “reducing hand-overs by avoiding certain tasks,” can reduce over-centralised decision-making and bureaucratic processes.

South African public healthcare also suffers from inadequate investments in training (Agyepong et al., 2017), insufficient IT investment and support (Buh et al., 2015), and a lack of a relevant BPM methodology and standards (Künzle & Reichert, 2011). All of these challenges were experienced in the project. Practice 1, “improving staff training,” addresses the inadequate investment in training. Practice 5, “incentivizing use of existing mobile software by doctors,” Practice 6, “developing low-cost task automation solutions such as Microsoft Office applications,” and Practice 8, “capturing data at source electronically rather than later,” all address increased use of IT.

This university collaboration was itself useful in terms of countering some of the identified challenges. A lack of BPM expertise has been identified as a major threat for public sector BPM initiatives (Valenca et al., 2013). The project was able to provide necessary BPM expertise that was useful. In terms of training, staff members learnt some methods and techniques of process redesign while participating in the project. In addition, this paper prescribes a useful BPM method and 10 best practices that can be used in hospitals, thereby contributing towards the lack of BPM methods for public healthcare.

The low-cost redesign steps prescribed were found to be appropriate in a low-cost, low-skill, and low-resource environment such as in the South African public hospital. Costing each step, while required in the BPTrends method (Harmon, 2014), was not found to be appropriate but enforcing reporting on KPIs to ensure transparency and accountability was seen as vital.

6 | CONCLUSION

While healthcare systems of low- and middle-income countries suffer from inefficiencies and need reform, literature does not have an appropriate framework to address complex and poorly funded public healthcare environments. This paper describes a low-cost redesign approach that was successfully applied in a public academic hospital in a middle-income country. This approach is low-cost from many perspectives: This is low-cost because the redesign steps are relatively simple and explained, the steps do not require any technology other than Microsoft Office and free BPMN modelling software, and the analysis was done as part of a class project by university students avoiding the use of expensive corporate consultants. The method is also suitable for low-resourced settings and consequently can be described as a low-cost approach. The study has shown that BPM methods are suitable for hospital processes as stakeholders confirmed that the recommendations are valuable. In contrast to process improvement methods, process redesign using external sources is more likely to overcome “silo” working pervasive in hospitals. We were also able to distil best practices to improve outcomes. Many of these practices can be workshopped with hospital staff (management, administration, nursing, and clinicians) enabling them to apply them as a toolbox. Yet through the study, it became apparent that resistance to change in a hospital setting must be tackled and overcome by designated project managers.

This study has limitations in that only four out of 11 cases were highlighted and discussed. In addition, the full BPM method employed could not be described. We acknowledge that not all universities have BPM courses and infrastructure to support such projects. Further longitudinal research is needed into the actual implementation of such initiatives.

ACKNOWLEDGEMENTS

We acknowledge the participation of hospital staff and are grateful for the input of the 55 IS students who worked on these projects.

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How to cite this article: Seymour L, Mwalemba G, Weimann E. Applied business process management: An information systems approach to improve service delivery in public hospitals of low- and middle-income countries. *E J Info Sys Dev Countries*. 2019;85:e12098. https://doi.org/10.1002/isd2.12098

APPENDIX A1

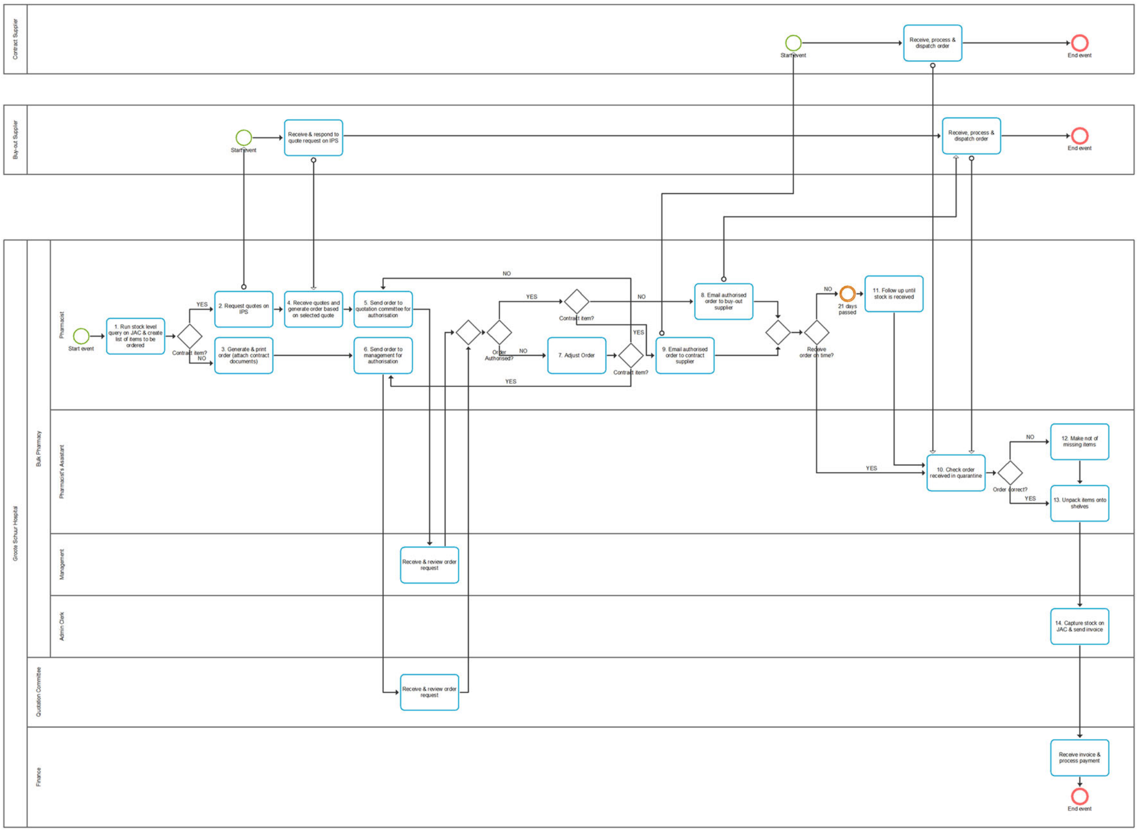

FIGURE A1 BPMN for bulk procurement (adapted from D1)